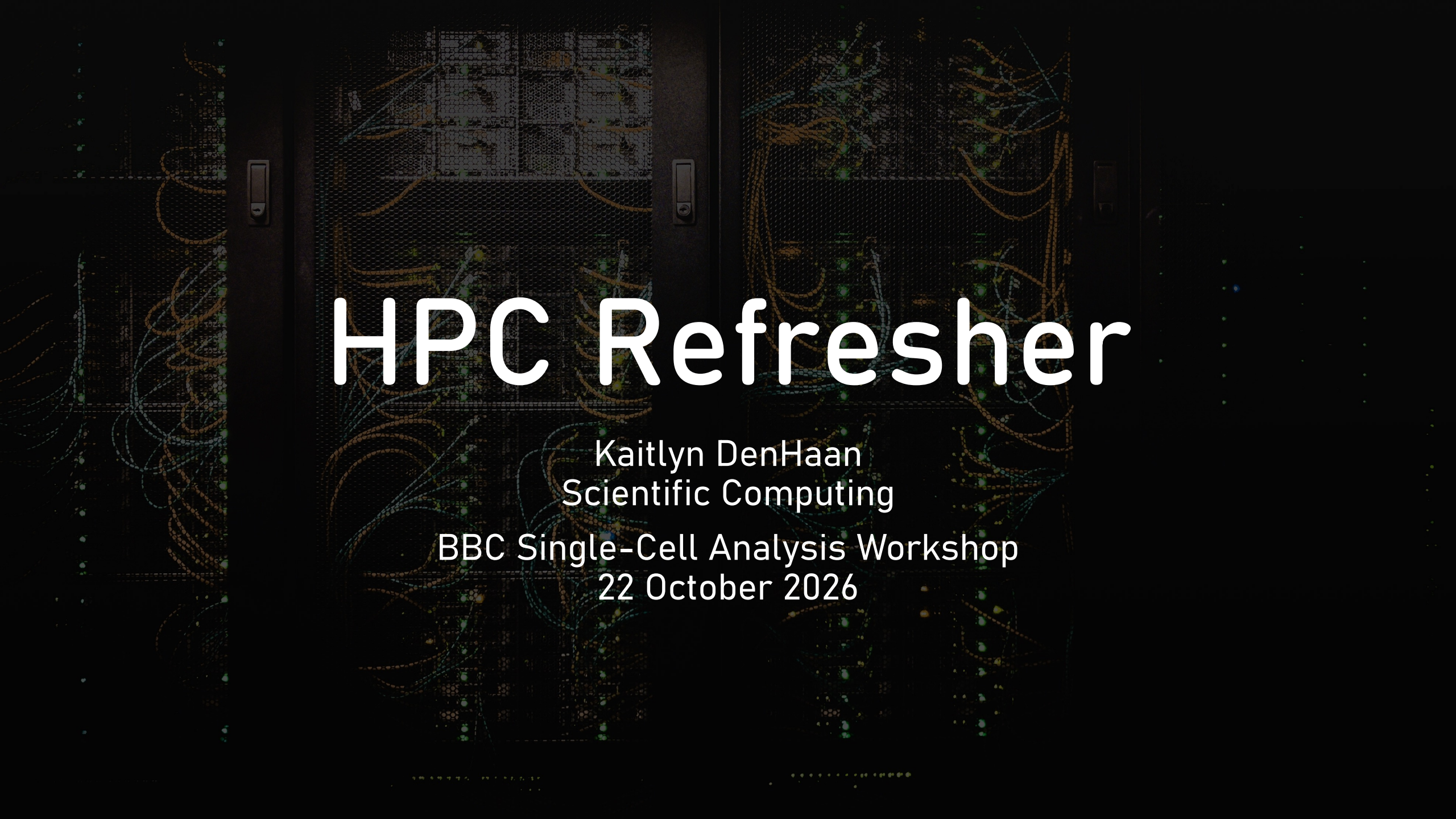


HPC Refresher

Kaitlyn DenHaan
Scientific Computing

BBC Single-Cell Analysis Workshop
22 October 2026

HPC Basics

VAI's HPC, when/why to use it, spec overview

What is HPC?

- VAI has a “high performance computing” cluster — a cluster of computers that you can run analyses on.
  - faster / more powerful than your laptop
- When you access the HPC from your laptop, you’ll be put on the submit/login/head/scheduler node.
- When you “submit” jobs, they’ll run on the “compute” nodes — these are the computers that do all the work for you.
- The HPC is also where all of our data is stored. There are folders for each lab, and shared folders that everyone can access.



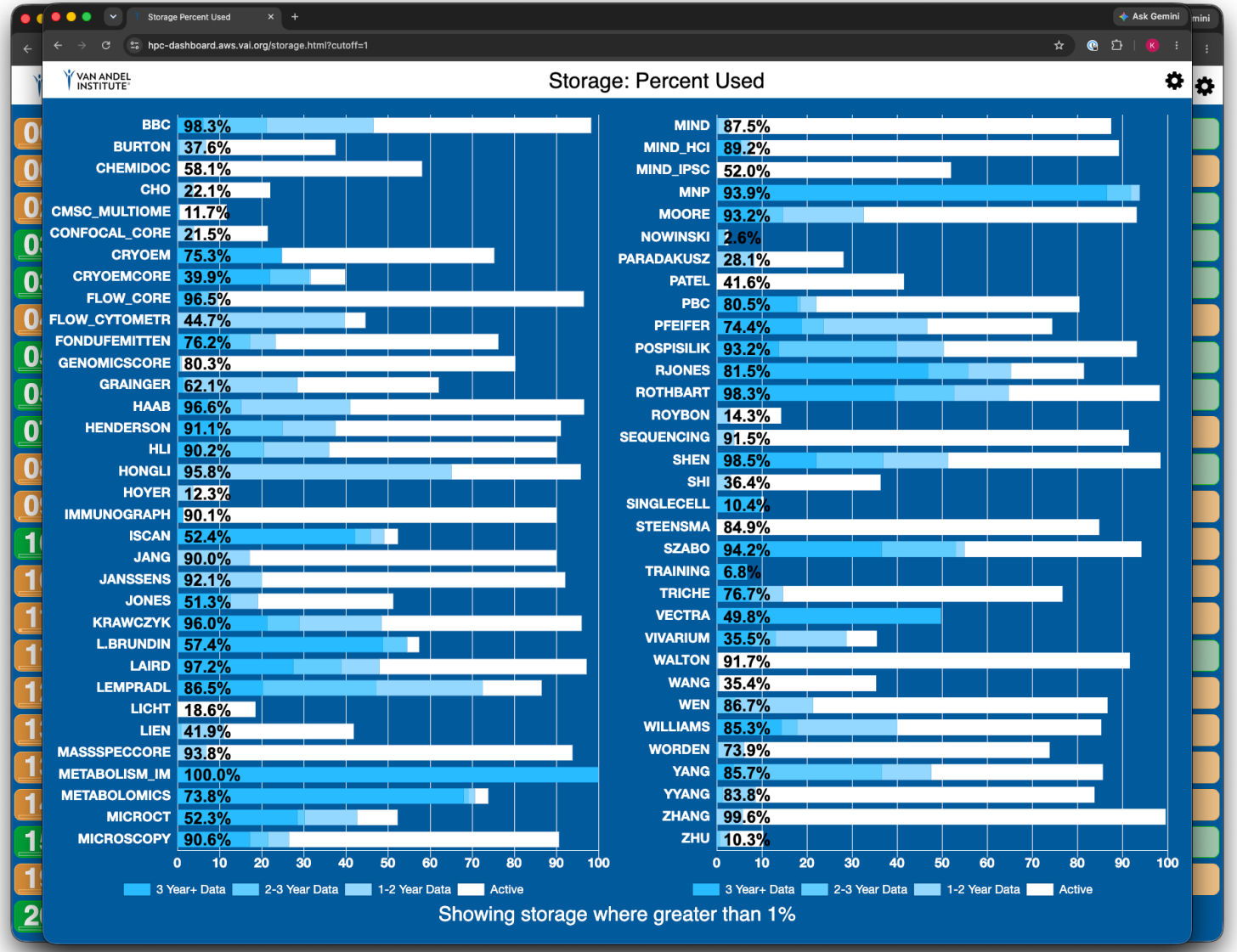
image from https://hpc.rtu.lv/hpc/introduction-to-hpc

Why use HPC?

- Go beyond your laptop's resources/capabilities
- Dramatically speed up your analysis
- Work directly on the files where they are
- Harness resources available to you
  - what *are* those resources?

Components of HPC

- **Computing Capacity** – 8376 Processing Cores, 89 Nvidia GPU cards
- **Storage Space** – 12.5 Petabytes (raw) with multi-site replication for data safety & disaster recovery
- **Network Speed** – 40 Gigabit / sec to each server; dedicated high speed to compute & storage

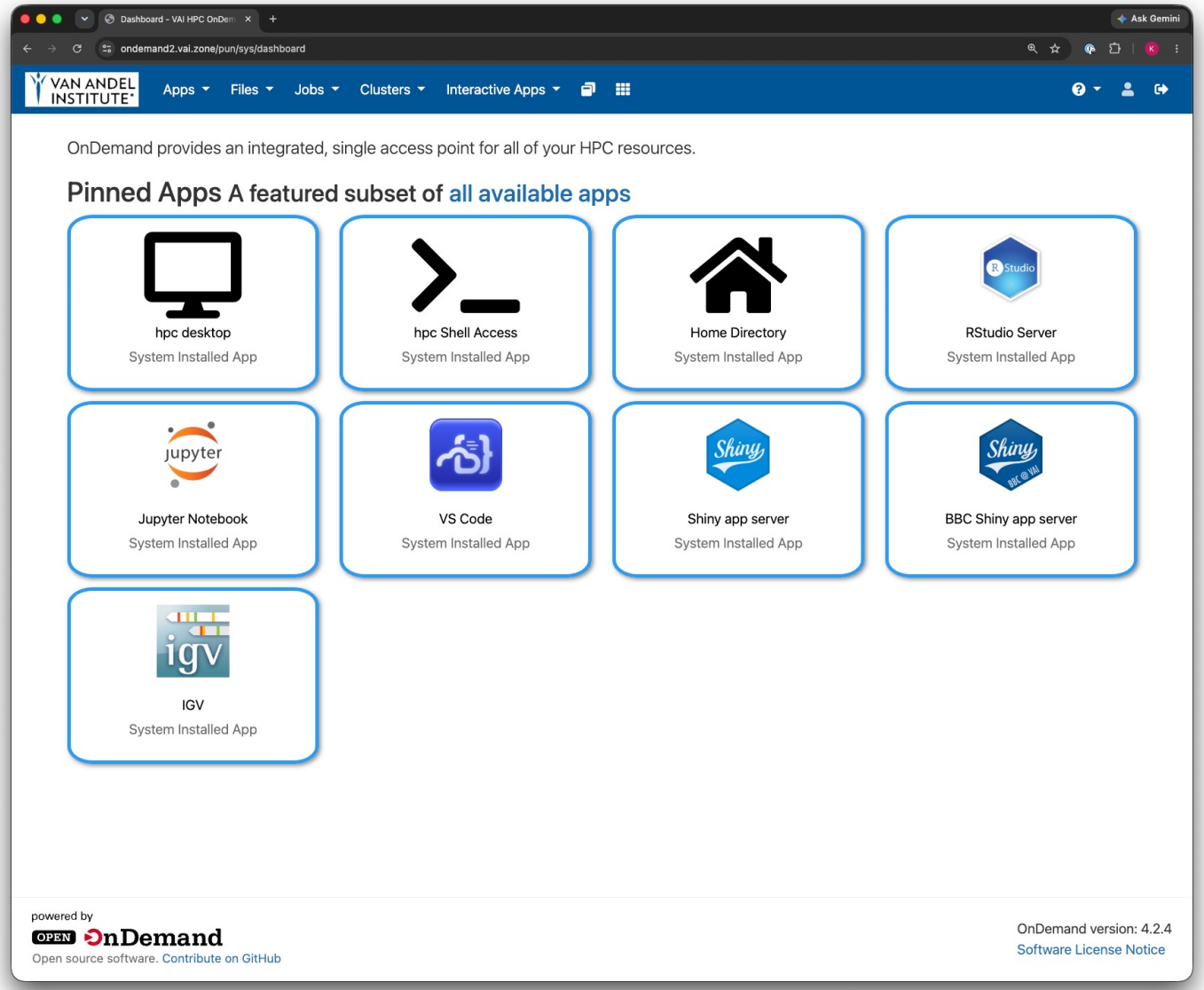


HPC OnDemand

use cases, available apps, how to connect

OnDemand

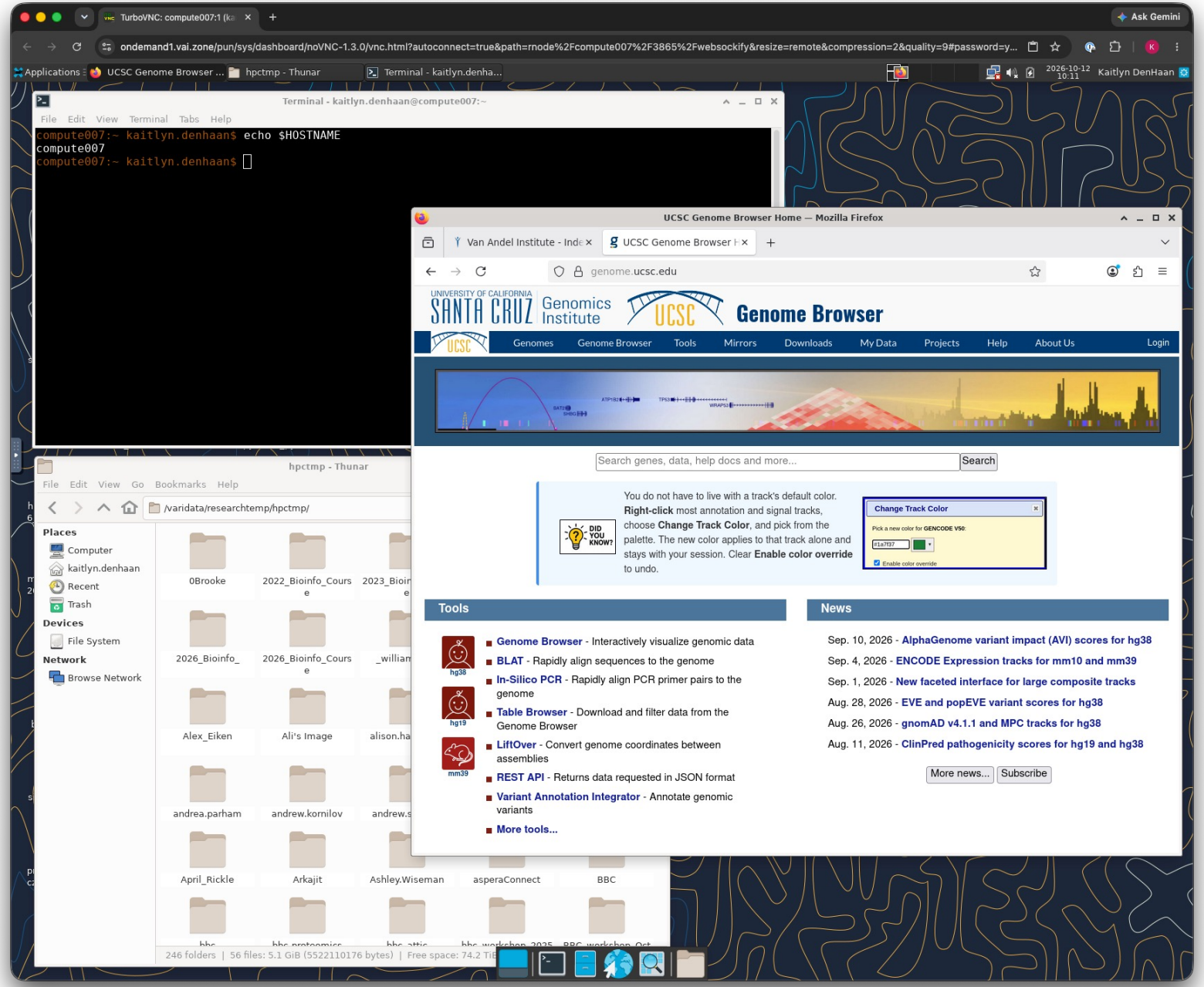
- Easy, web-browser based way to connect to & use the HPC
- Follow along:
  - Apps available



HPC Desktop

Desktop computer style interface to open multiple apps…

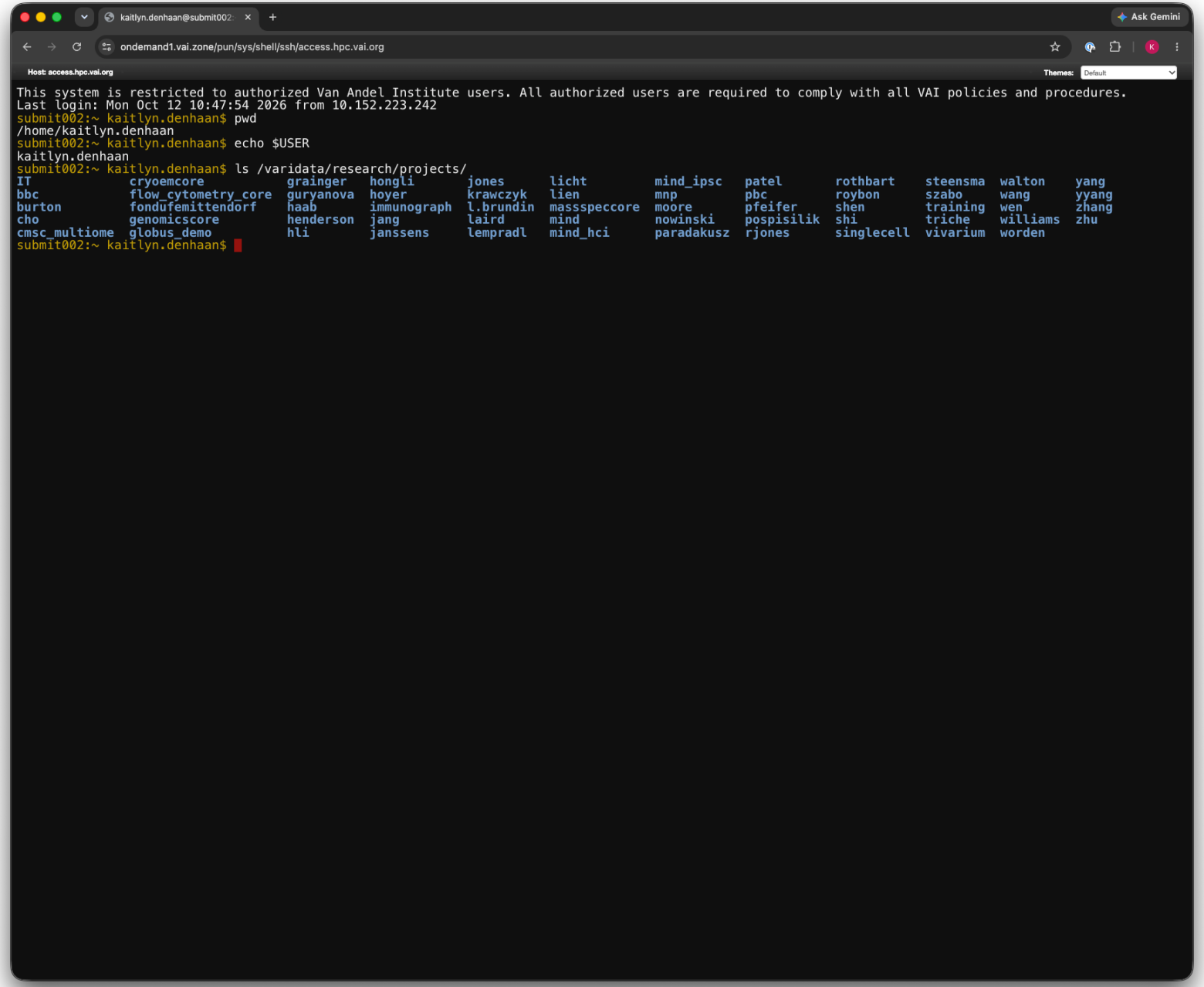
- see files/folders
- browse the web
- use a terminal (already on a compute node)
- launch other interactive GUI applications



Shell Access

Terminal-style interface to access HPC

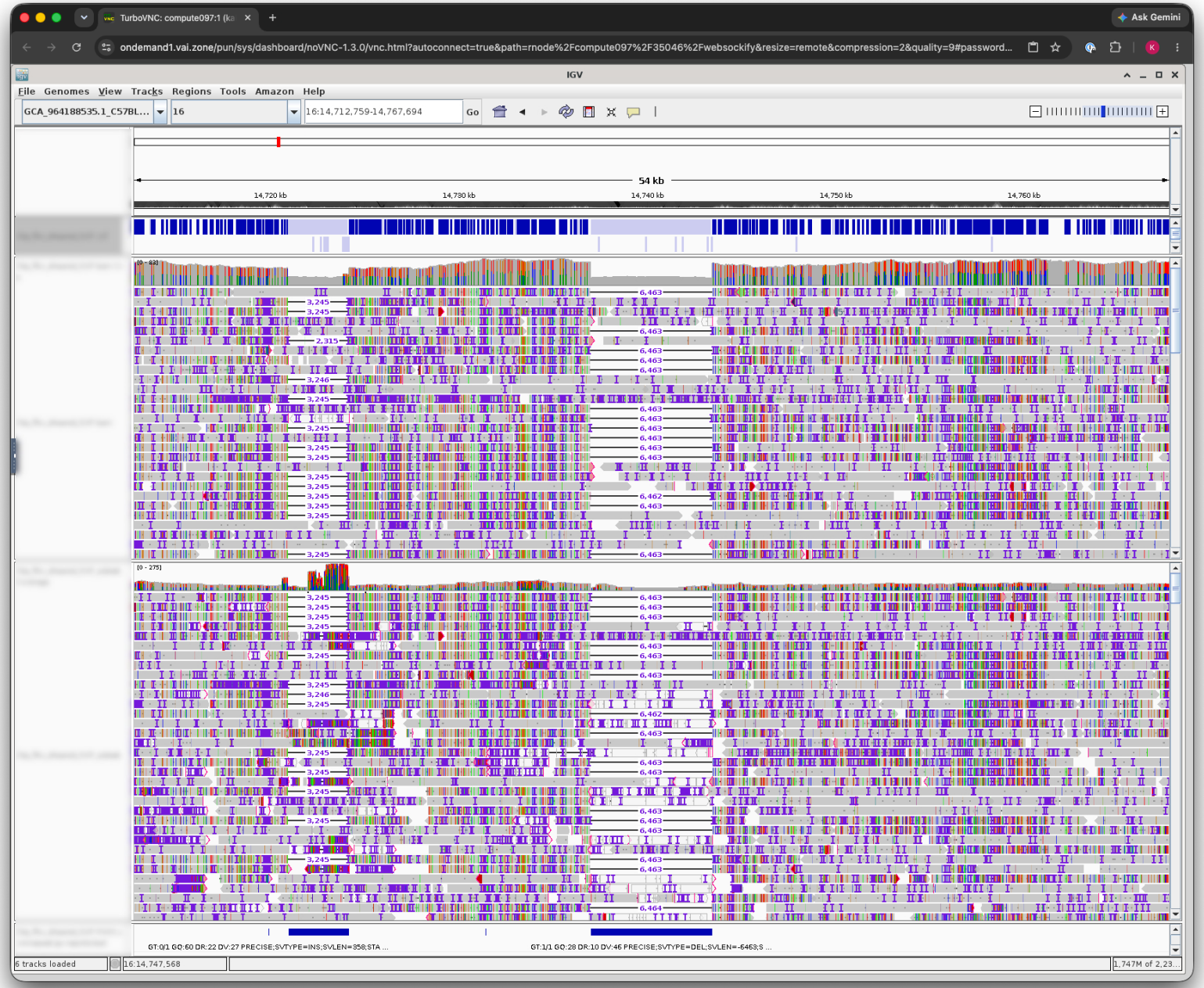
You will be connected to the submit node.



IGV

Load & view genomic data

Note: HPC nodes have much more memory than your laptop – useful when viewing large data files!

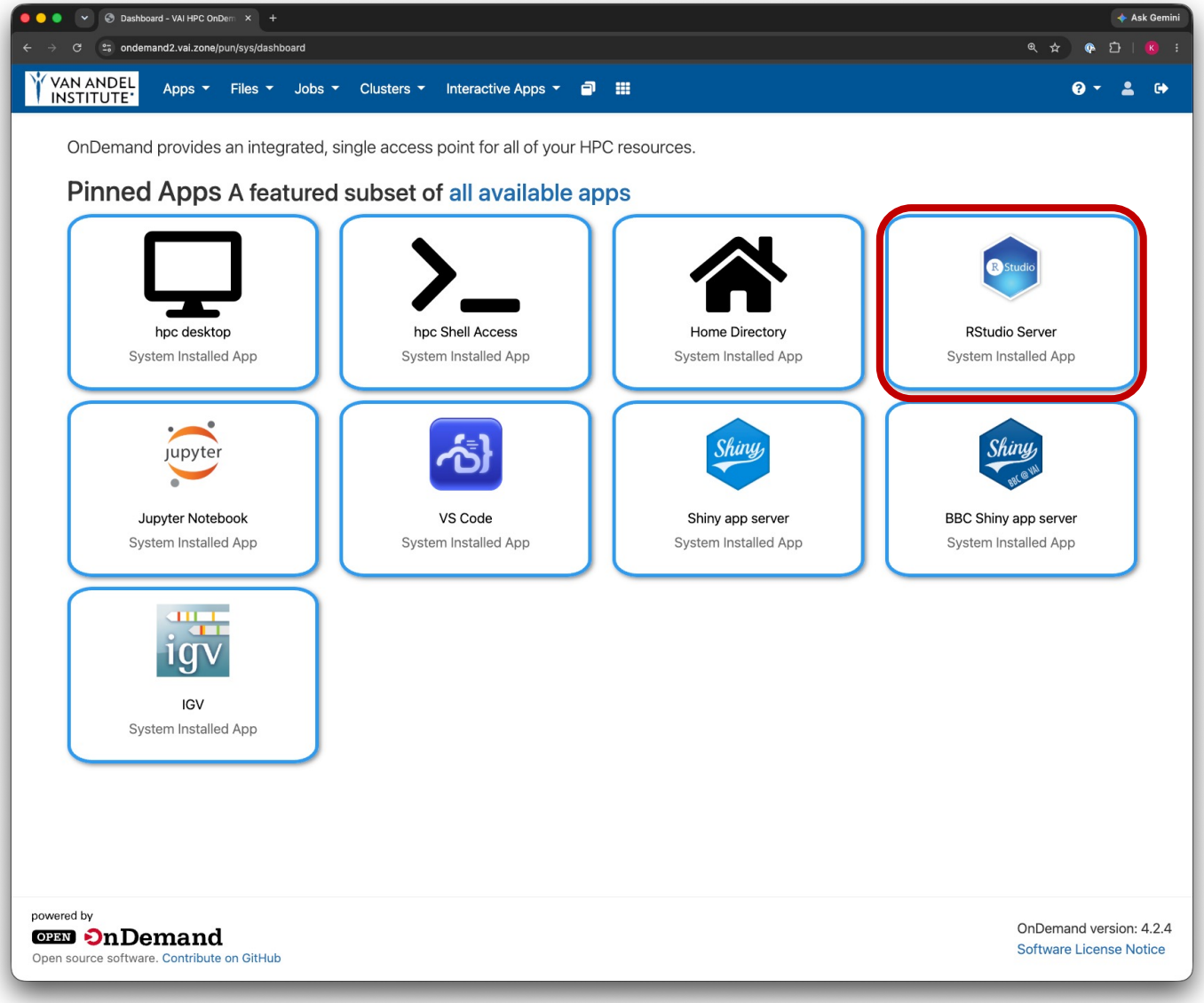


Popular Apps

RStudio – interactive environment for programming in R

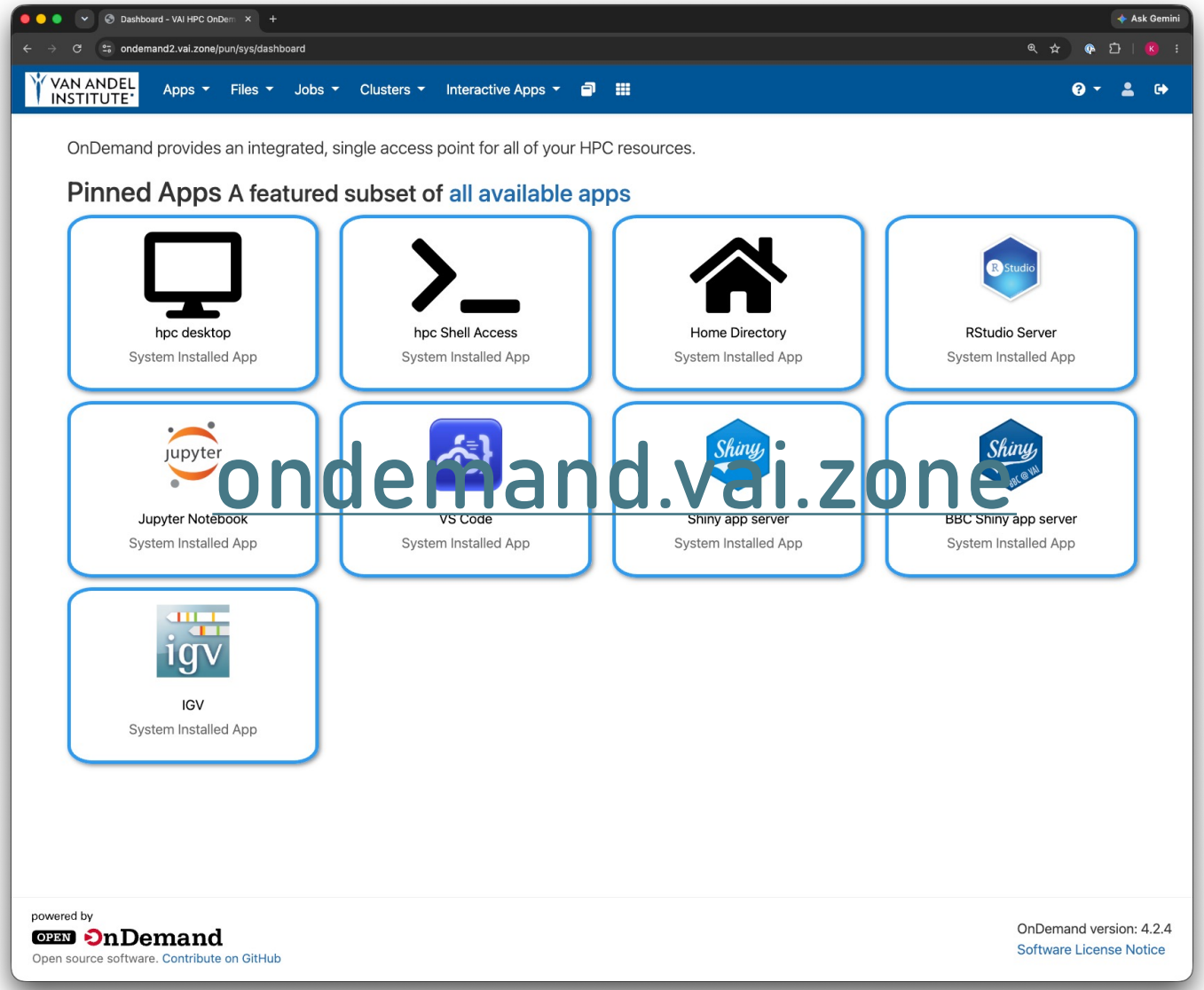
Jupyter – create computational documents primarily in python

VS Code – all-purpose code editor, with a built-in terminal and multiple extensions



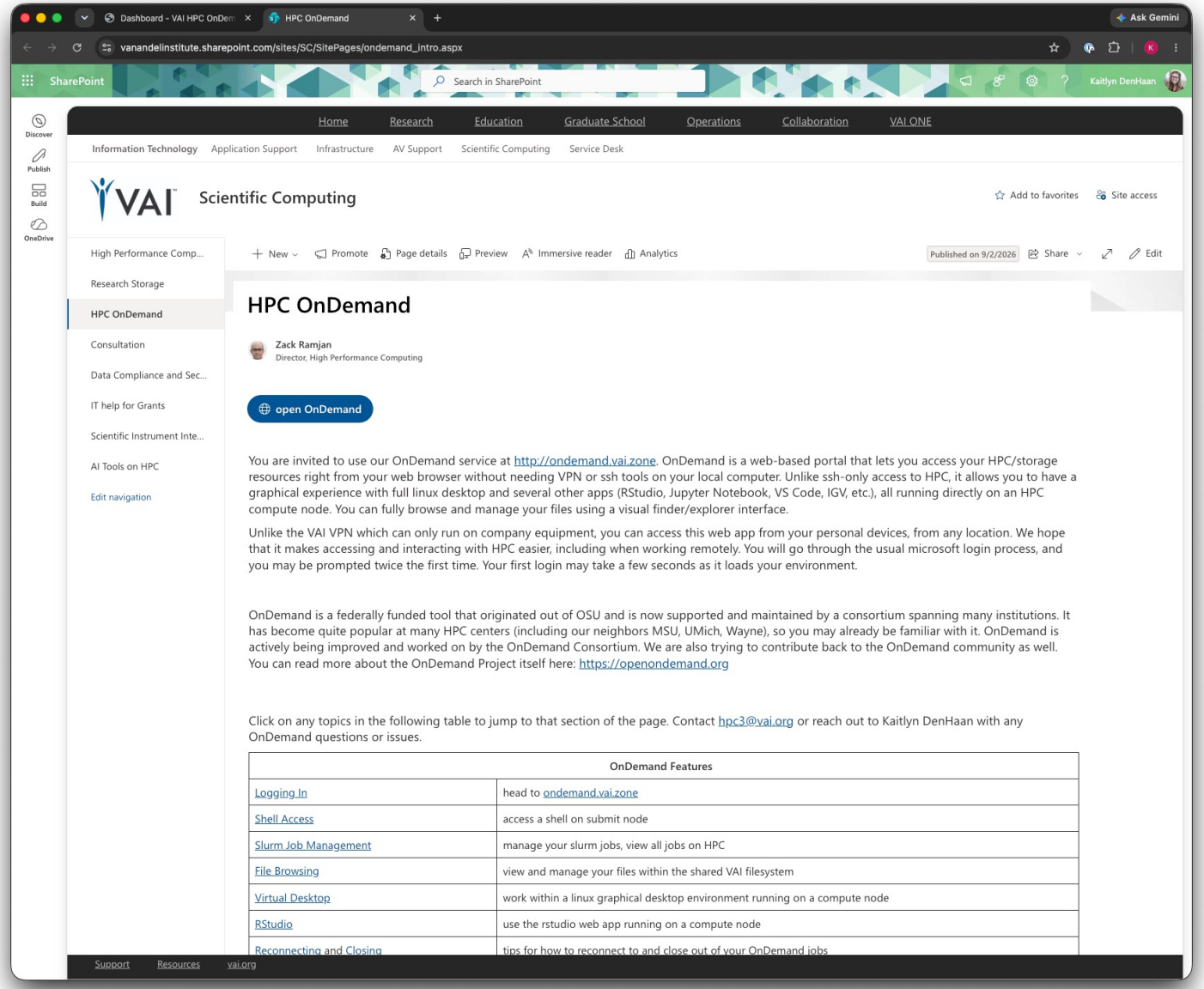
OnDemand

- Easy, web-browser based way to connect to & use the HPC
- Follow along:
  - Apps available
  - Signing in



OnDemand

- Easy, web-browser based way to connect to & use the HPC
- Follow along:
  - Apps available
  - Signing in
  - Launch form
- Finding help?
  - OnDemand Sharepoint page!
  - Reach out to Kaitlyn



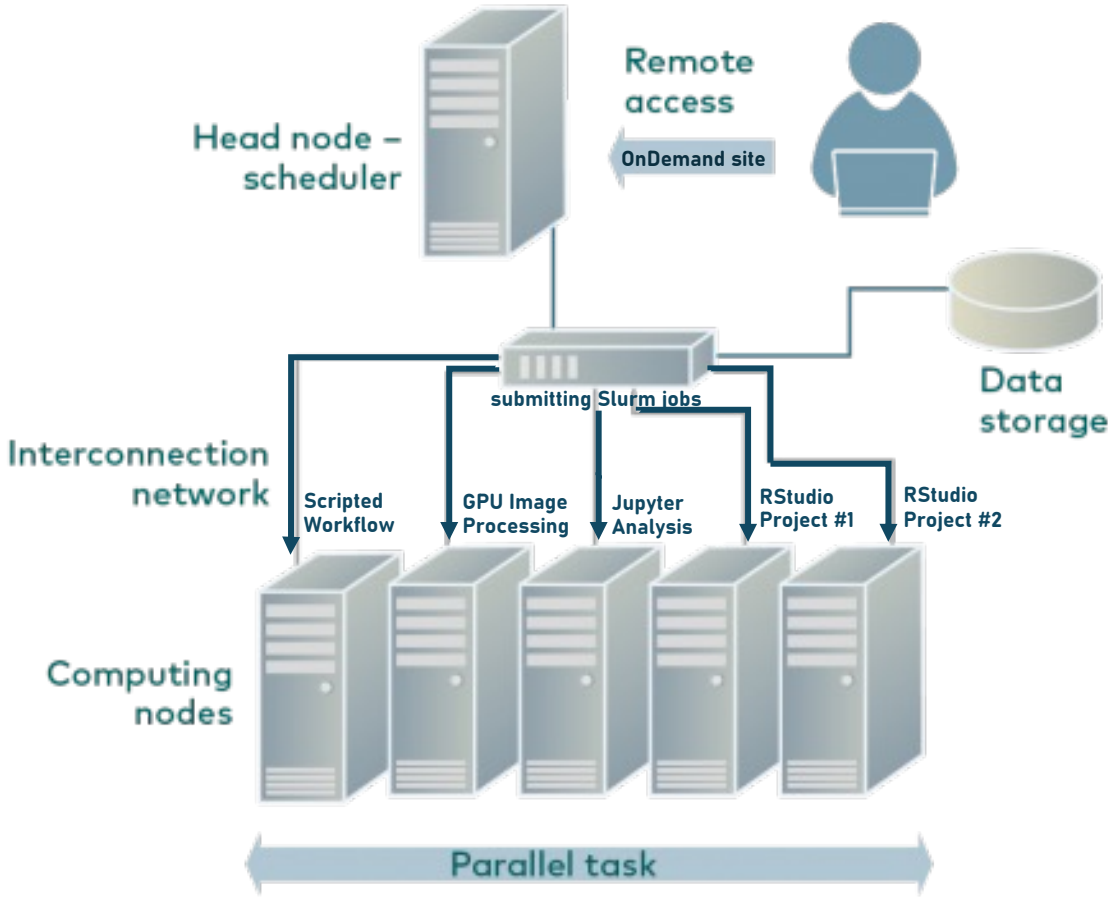
Slurm

job scheduling, and available resources

Slurm

- System that schedules all jobs
- Submit a slurm job when you want to run something on a compute node
- Slurm handles the line/queue and your job will automatically begin when it's first in line & resources become available

- What partitions are available to submit to? & when to use each?



partition	purpose	# of nodes available	specs of each node	max limit: run time	max limit: cores per user	max limit: # of running jobs	max limit: # of queued+running jobs	other notes
short	small/medium jobs	30	16 cores, 64 GB memory	12 hours	64	64	400	• most OnDemand jobs fit well here • no MPI jobs that use more than one node
long	most long-running intense jobs	22	64 cores, 500 GB mem	96 hours (4 da	6	100	400	• works well for scripted jobs that use >16 cores • multi-node MPI jobs OK
bigmem	big jobs needing high cpu & 1TB mem in a single node	2	128 cores, 1 TB memory	168 hours (7 days)	256	1	0	• only used for jobs with extreme memory needs • multi-node MPI jobs OK
gpushort	small jobs that need gpu	8	1 gpu NVIDIA L40S, 48 GB v 48 cor 90 GB mo	24 urs	ode	1	8	• works well for dorado basecalling or smaller tests of tools utilizing a GPU
gpulong	big or long-running jobs that need gpu	4	4 gpu NVIDIA L40S, 48 GB vram, 192 cores, 768 GB memory	168 hours (7 days)	1 node	4	8	• works well for structural biology & image analysis applications that utilize multiple GPUs

See Sharepoint!

Choosing Resources for Slurm Jobs

Not sure what resources (cpus/gpus, memory, time limit, etc.) to request? It can be difficult to know how much to request, especially if you're newer to high-performance computing – request too few and your job might fail by hitting its limits, request too many and you're leaving resources unused.

A few things to keep in mind as you determine an appropriate amount to request for your jobs:

- **getting started** – when running tests to determine how many resources you'll need, request a reasonably modest amount, and if you run into a particular limit (e.g., memory, runtime), submit a new job with that limit increased. When running production-level code for formal analysis, request an amount *slightly higher* than what you think you'll need to give your job a bit of extra buffer.
- **check tools' documentation** – most programs use a single cpu unless explicitly configured otherwise. Look up the documentation for the tools you're using to see if there's a threads/cores/workers option. For some software, you'll often see an option like "-t 8" to use 8 threads. For R functions, check whether there's a "BPPARAM" parameter and familiarize yourself with BiocParallel. If you utilize these options, request that number of cpus from Slurm.
- **RStudio OnDemand** – most R commands use a single cpu, so for interactive code development and testing, 1 cpu is usually sufficient. If you decide to scale up your code with using parallel or multithreaded code later on (see point above), you can increase your cpus requested accordingly. Simply requesting additional cpus will not make ordinary R code run faster.
- **large workflows** – test your workflow with scaled-down input files and monitor the resources they use. When you're ready to resubmit your workflow with your full dataset, scale up. When possible, determine resources using previous runs on real data of similar size.
- **single-user public nodes** – when running jobs on a public node (bigmem/long/etc.) or private lab-owned nodes), no other user will be able to run jobs on this node at the same time as you. This means you can make use of all the cpus available on this node – whether submitting several jobs to this same node or allowing a parallel job to use more of the node's resources. You should still request cpus and memory that accurately reflect what each job will use, but using only a fraction of the cpus on that node doesn't necessarily help other users.
- **GPUs** – GPUs should only be requested when using software that were written with GPU-acceleration in mind. Check your tool's documentation for any GPU-related information and submit jobs to an appropriate partition accordingly. If you're unsure whether the tool you're using supports GPU acceleration, it probably doesn't, and won't benefit from being run on a GPU partition.
- **need more guidance?** – you can always reach out to the HPC team (e.g., stop by our monthly office hours, send us a quick email, etc.), and we'd be happy to work with you!

See Sharepoint!

Best Practices

how be a good HPC user!

Best Practices – Pro Tips for HPC Usage

- don't work on the same file in multiple locations
  - e.g., an excel file located w/in the VAI filesystem: don't edit this on your computer while *also* trying to open it elsewhere.
    This confuses the filesystem and can lead to annoying locks and potentially even losing the file entirely.
- use a partition that meets (but doesn't exceed) your needs
  - e.g., don't use a "long partition" node for a single-cpu RStudio job that only needs to run for 2 hours.
    If you run into limits (# cpus, memory, time limit), first increase any self-imposed limits, and use the next largest partition if you've hit the max for that partition.
- close out of your jobs when done using them – including OnDemand
  - let's practice doing this together!

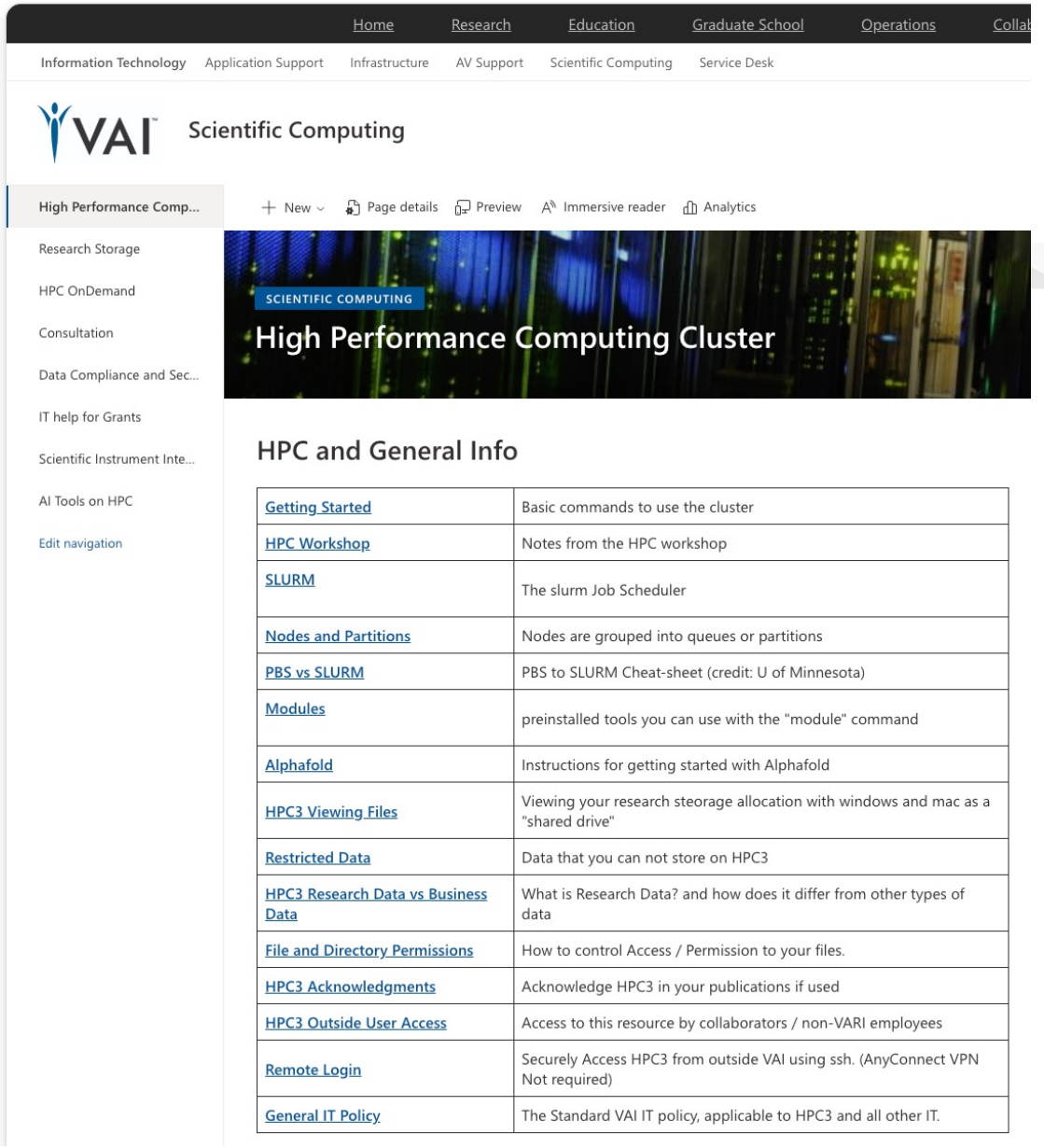
Reaching out for help

Who to contact & how to reach us!

SharePoint Pages for Resources & Documentation

Great resource to look through, includes general information, step-by-step tutorials, and links to past presentations from our group

- how to transfer data externally w/ Globus
- our free cloud archival storage solution & how to move files up to the cloud
- OnDemand apps overview
- & more!



HPC Team

- Zack Ramjan
- Jason Kotecki
- Kenneth Mendenhall
- Kaitlyn DenHaan

Best way to contact us:

email hpc3@vai.org: creates a ticket for all of us to see – you get the quickest & most informed response

How to Contact Us

Some options:

- **email** hpc3@vai.org (creates a ticket for any one of us to take – whoever can get you the quickest and most informed response)
- **office hours** – in THIS room! monthly (2nd Wednesday @ 3pm). come & chat, ask your questions, very casual environment (HPC, BBC, OIC, Office of the Cores)
- **any other random things?** – reach out to me (or anyone on the team) directly, always happy to chat, can get you connected to whoever should be involved

Session Overview

HPC isn't scary; it's a resource that's here for you to use to make your work faster, easier, and more streamlined.

OnDemand is a useful tool for connecting to VAI's HPC in an interactive, web-based format, with multiple apps available.

Feel free to reach out or come chat with us about any issues or questions you have!

Enjoy the rest of the workshop!

Next Up: Single-cell library preparation and sequencing with Katelyn Becker